SAMEER KHAN
San Francisco, CA | sameerikhan3@gmail.com | LinkedIn

EDUCATION

University of California, Berkeley - *B.S. Mechanical Engineering*

Relevant Coursework: Engineering Project Management, Advanced Engineering Design Graphics, Dynamic Systems and Feedback

SKILLS

CAD: OnShape, AutoCAD, SolidWorks, Creo, REVIT, DFM| 3D Modeling, Assemblies, Simulations
Programming: Python, MATLAB, Linux, MicroPython | Embedded Systems, Algorithm Development
Engineering: GD&T, Control Systems, FEA, Product Design, Simulation, Modeling, Surface Modeling
Software : MS Office Suite, MS Project, LabVIEW, Simulink, Arduino IDE
Soft Skills: Communication, Project Planning, Technical Writing, Initiative, Adaptability, BOM

EXPERIENCE

John Bertoldi Inc, JBI San Francisco, CA
Assistant Project Engineer Dec 2023 – Aug 2024

- Reviewed RFIs, drawings, and field issues to ensure alignment with design intent and project requirements.
- Conducted regular site walks to monitor progress and verified contractor progress against the project schedule.
- Helped track project milestones and updated progress reports to assist the project manager in overall scheduling and resource planning.
- Gained hands-on experience in technical documentation and project tracking.

University of California, Undergraduate Research Apprentice Program (URAP), Berkeley, CA
Design Engineer Intern Jan 2023 – Dec 2023

- Designed a peritoneal dialysis machine aimed at treating infants up to 9 months.
- Developed a pinch actuation system using a linear actuator and a graphical user interface.
- Drafted components using SolidWorks and GD&T principles and integrated a microcontroller and actuator (IOT).
- Calibrated sensors (force, pressure, displacement) to ensure accurate system feedback and reliable test data.
- Documented results and performance data in compliance with regulatory standards, contributing to the Device History Record.

PROJECTS

Robotic Writing Arm - Project Lead

- Designed and assembled a gantry robot for automated writing, achieving 95% accuracy in replicating complex images.
- Conducted hands-on testing of mechanical components and assemblies, adjusting design to optimize performance
- Developed detailed testing protocols for validating system functionality and troubleshooting performance issues

Automatic Plant Watering System

- Built a smart irrigation system using an ESP32 microcontroller and a capacitive moisture sensor to maintain optimal soil hydration in plants at a desired moisture level.
- Developed an app to visualize moisture levels in real time, enabling user-friendly remote monitoring.
- Designed and assembled housings and circuits, integrating sensors and embedded software for standalone operation.